

6.6.1 Staking

Staking is an important component of the Network design to incentivise appropriate behaviors from Network participants such that they can be rewarded for long term alignment, and have stakes slashed as punishment for undesirable behaviors such as not fulfilling node uptime requirements or not adhering to user voted requirements on user experience.

Staking rewards in the Network are derived from the Reward pool. One can understand given a total rewards budget $R_G(t)$, a staking rewards budget $R_S(t)$ as the following:

$$R_s(t) = W_s R_G(t)$$

W_s is set by governance, with a proposed predefined range of 0.25-0.66.

Calculating the rewards Each staker i possesses a stake-weight $w(i, t)$ and a consequent share of the overall sum of stake-weights:

$$s_{stWeight}(i, t) = \frac{w(i, t)}{\sum w(j, t)}$$

We'll denote the total sum of stake-weight as ΣW and the total sum of staked tokens as ΣS .

To counter the concentration of such a stake-weighted reward scheme, there is a cap c_{MAX} imposed on the share of stake-weights (proposal: $c_{MAX} = \frac{1}{10}$), hence the effective share of stake-weights is:

$$s_{stWeight}(i, t) = \min\left(\frac{w(i, t)}{\Sigma W}, c_{MAX}\right)$$

For each staker i , the individual rewards are their stake-weight share multiplied with the staking rewards, which are a part of the general rewards budget, that is:

$$R_S(t) = W_S * R_G(t)$$

As mentioned earlier, W_S is set by governance. The individual staking rewards are then:

$$r_S(i, t) = s_{stWeight}(i, t) * R_S(t)$$

Fixed Staking The Fixed staking system allows participants to lock up their tokens for predetermined periods. This approach provides flexibility while providing for network stability and long-term commitment from participants.

Key features of the fixed staking system include:

1. **Flexible Duration:** Users can choose staking periods ranging from 1 week to 2 years (104 weeks), the duration users can stake are measured in weekly increments.